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Novel Class of Retarded, Newtonian, Single-Phase HCl-Based Stimulation Fluids - A Laboratory Characterization

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Abstract

The current industry incumbent compositions of retarded acids have certain drawbacks: (1) environmentally harmful, (2) unstable at high temperatures or shear rates, (3) excessive frictional pressure losses due to high viscosity, (4) high-cost additives, (5) complicated mixing procedures, etc. This work reports the development and laboratory characterization of the novel class of retarded acid that contains 15 wt% HCl but overcomes all the disadvantages mentioned above.

A rotating disk apparatus (RDA) was used to determine the dissolution rate of calcite-marble in the novel class of retarded acids and 15 wt% HCl solution. Experiments were performed at a temperature of 250°F, disk rotational speed of 587 rpm, and pressure of 1,150-1,300 psig. The dissolution rate of calcite-marble was calculated from the calcium ion concentrations in the samples of the solutions withdrawn during the RDA tests. Corrosion rates of Cr13, L80, P110, and T95 steel coupons have been measured using an autoclave at a temperature of 250°F, and a pressure of 450 psig, for 6 hours. Kinematic viscosity was measured using a capillary viscometer at a temperature of 73-200°F.

The novel class of retarded acids contains 15 wt.% HCl, and is a single-phase, Newtonian, low-viscosity, i.e., water-like liquid. The acid is adjustable, i.e., has the capability to fine-tune the reactivity from 1.2 to 9200 times slower than 15 wt.% HCl at 250°F. Besides tunable reactivity, no significant corrosion of steel coupons is observed without adding any corrosion inhibitor and/or corrosion inhibitor aid. The measured corrosion rates were 0.00001 to 0.05061 lb/ft² in 6 hours. Moreover, novel acid packages act as an H₂S scavenger and can be used in sour wells.

This work reports the development of innovative retarded acid packages that exhibit tunable reactivity, low corrosion rate on many steel grades, high-temperature stability, and multifunctionality. Such acid packages are of great interest for the oil and gas industry, CO₂ storage, and geothermal projects around the world.

Introduction

The global energy sector relies heavily on efficient hydrocarbon extraction from carbonate reservoirs, which account for approximately 60% of the world's oil and gas reserves [1]. Matrix acidizing, a key

stimulation technique, enhances productivity by creating highly conductive channels, or wormholes, through the dissolution of carbonate rock. Hydrochloric acid (HCl) is the most widely used stimulation fluid due to its high dissolving capacity, wide availability, and low cost. However, its rapid reaction rate with carbonate formations, particularly at high temperatures ($>200^{\circ}\text{F}$), results in near-wellbore spending, limiting deep penetration and reducing stimulation efficiency [2]. This challenge is exacerbated in high-pressure, high-temperature (HPHT) reservoirs and long horizontal wells, where suboptimal injection rates lead to inefficient wormhole formation [3]. To address these limitations, retarded acid systems have been developed to control the acid-rock reaction rate, enabling deeper penetration and improved productivity of well. Traditional retarded systems, such as gelled acids and emulsified acids, have significant drawbacks, including high friction pressures, complex mixing requirements, and environmental concerns due to diesel-based formulations [4]. This has driven the development of novel retarded, Newtonian, single-phase HCl-based stimulation fluids, which offer a promising alternative for carbonate acidizing.

The novel class of retarded, Newtonian, single-phase HCl-based stimulation fluids represents a significant advancement in acidizing technology. Unlike traditional retarded systems, these fluids are polymer-free and diesel-free, maintaining a low-viscosity, single phase profile that simplifies mixing and pumping operations while reducing environmental impact [6]. These fluids achieve retardation through innovative chemical retarders that slow the acid-rock reaction by 1.2 to 10.7 times compared to 15 wt. % straight HCl at 250°F , as demonstrated in laboratory studies [3]. The Newtonian behavior ensures consistent viscosity under varying shear rates, facilitating high- rate or coiled-tubing operations without the shear-thinning or shear-thickening complexities of non-Newtonian fluids [5]. Laboratory characterizations, including static carbonate dissolution tests, reaction kinetics measurements, and coreflow testing on Indiana limestone cores, have shown that these fluids produce deeper, more efficient wormholes with pore volumes to breakthrough (PVBT) as low as 0.41 at optimal injection rates [7]. Computed tomography (CT) scans reveal single, dominant wormhole trajectories with minimal branching, indicating effective acid placement [4]. Additionally, these fluids exhibit compatibility with corrosion inhibitors, meeting industry standards for corrosion rates on L80, P110, T95, and 13Cr steel grades without additional inhibitors [3].

Commercially, these novel fluids are gaining traction for their operational simplicity and performance in HPHT carbonate reservoirs. Field applications in regions like Kazakhstan and offshore Sarawak have demonstrated production increases of up to 25% compared to organic acid treatments, attributed to improved wormhole efficiency and higher dissolution power [4, 5]. Companies such as SLB and ChampionX have pioneered these systems, with products like Single-Aqueous-Phase Retarded Acid (SAPRA) showing reduced friction pressure and compatibility with acid friction reducers [5]. The ability to mix and pump these fluids on the fly reduces surface equipment requirements and operational downtime, enhancing cost-effectiveness. The systems are particularly suited for long horizontal wells and reservoirs with temperatures up to 300°F , where traditional emulsified acids struggle with high friction and sensitivity to reservoir conditions [6]. Advanced simulation software has further optimized treatment designs, enabling precise fluid volumes and pumping rates, with reported acid volume reductions of up to 44% in some fields.

Despite their promise, these novel fluids face several challenges. The retardation mechanism is highly dependent on injection rate and reservoir conditions, requiring precise optimization to avoid near-wellbore spending [5]. Compatibility between retarder chemistries and corrosion inhibitors must be carefully tuned to prevent molecular interference, as the retarder binds to positively charged carbonate surfaces while inhibitors target negatively charged metallic surfaces [5]. In low-concentration brines, such as those containing trace metals like manganese or lithium, competing ions (e.g., magnesium, calcium) can reduce retardation efficiency, necessitating pre-treatment or tailored formulations [8]. Scaling these systems to diverse reservoir chemistries remains a hurdle, as variations in salinity and mineralogy affect reaction kinetics. Environmental concerns, while reduced compared to diesel-based emulsified acids, persist due to the potential for residual chemical discharge. High capital costs for developing and testing these formulations also pose economic barriers, particularly in price-sensitive markets [11]. Looking ahead, the

development of novel retarded, Newtonian, single-phase HCl-based fluids is poised to transform carbonate acidizing. Advances in retarder chemistry, such as nanostructured additives or hybrid organic-inorganic molecules, could further enhance retardation efficiency and selectivity, particularly in low-concentration brines [9]. Integration with renewable energy sources, such as solar-powered pumping systems, could reduce the carbon footprint of acidizing operations. Machine learning and computational fluid dynamics (CFD) are emerging as tools to optimize acid placement and predict wormhole propagation, improving treatment outcomes [10]. Regulatory incentives for low-impact stimulation fluids may accelerate adoption, especially in environmentally sensitive regions. As industry shifts toward sustainable practices, these fluids offer a pathway to balance high performance with reduced environmental and operational footprints, potentially setting a new standard for carbonate reservoir stimulation.

Table 1—Comparison of HCl-based stimulation fluids for carbonate reservoirs

Fluid Type	Key Characteristics	Challenges
Straight HCl	High dissolving capacity; low cost	Rapid reaction; near wellbore spending
Gelled acid	Increased viscosity; moderate retardation	High friction; polymer residue
Emulsified acid	Deep penetration; diesel-based	High friction; environmental concerns
Novel retarded single-phase	Low viscosity; $1.2 - 10.7 \times$ retardation; diesel-free	Injection rate sensitivity; high development costs

The objectives of this study are to: (1) quantify the dissolution performance of tunable retarded acid (TRA) systems using a Rotating Disk Apparatus (RDA) across a range of temperatures, (2) assess their corrosion resistance on common well tubular alloys (Cr-13, L- 80, P110, T95) without inhibitors, (3) compare TRA performance against emulsified and in-situ gelled acids, and (4) evaluate their operational and environmental benefits for carbonate reservoir stimulation.

Material

We have designed the TRA systems to be formulated with 15% HCl as the base acid, with selected tests conducted at concentrations up to 32% HCl to evaluate scalability. Organic acids, such as formic acid, were tested to assess compatibility with alternative acid types. Six proprietary TRA packages (Package 1 to Package 6) were developed, each incorporating a unique retarder formulation to achieve a spectrum of reaction rates. The retarders are chemically engineered to form temporary complexes with HCl, slowing their reactivity with carbonate minerals without compromising dissolving power. The formulations are compatible with viscosifiers to adjust fluid viscosity for specific applications, such as fracture acidizing. Calcite marble disks (99.5% CaCO₃, diameter 1.5 inches, thickness 0.5 inches) were used for dissolution rate measurements to simulate carbonate reservoir rock. The disks were polished to a surface roughness of <5 gm to ensure consistent surface area and reaction conditions. Corrosion tests were conducted on metal coupons (2 × 1 × 0.125 inches) made of Cr-13, L-80, P110, and T95 alloys, representing standard well tubular materials. The coupons were cleaned and weighed to a precision of 0.1 mg before testing. All TRA formulations were tested without corrosion inhibitors or intensifiers to evaluate their intrinsic corrosion resistance. The TRA systems are designed to be environmentally friendly, avoiding diesel or hazardous additives common in emulsified acids. The single-phase nature simplifies preparation and reduces waste, aligning with industry trends toward sustainable stimulation fluids [12].

Methodology

We have designed and laboratory tested a novel and effective barite dissolver that will dissolve a variety of barite samples, including natural barite rocks and oil industry barium sulfate scales, with much higher yield than current chelating systems.

Dissolution Rate Testing

Dissolution rates were measured using a Rotating Disk Apparatus (RDA), a standard tool for studying acid-rock reaction kinetics [3]. The RDA consists of a calcite disk mounted on a rotating shaft, immersed in 500 mL of acid solution within a pressurized reactor. Tests were conducted at temperatures of 180, 250, and 350°F, with a rotational speed of 500 rpm to ensure laminar flow and consistent mass transfer. The dissolution rate was determined by measuring the rate of calcium ion release into the solution via inductively coupled plasma (ICP) spectroscopy, expressed as mol/sec cm^2 . The equation for dissolution rate is:

$$J = \frac{A[Ca^{2+}] \cdot V}{A \cdot t}$$

where J is the dissolution rate, $A[Ca^{2+}]$ is the change in calcium ion concentration (mol/L), V is the solution volume (cm^3), A is the disk surface area (cm^2), and t is the reaction time (sec). Each TRA package was tested with 15% HCl at 250°F, with additional tests at 180 and 350°F for select packages. Straight 15% HCl, 5% HCl in-situ gelled acid, and 15% HCl emulsified acid were tested as benchmarks. Tests were conducted in triplicate, with post-test disk surfaces analyzed using scanning electron microscopy (SEM) to evaluate wormhole formation patterns. Figure 1 shows the schematic diagram of the rotating disk apparatus.

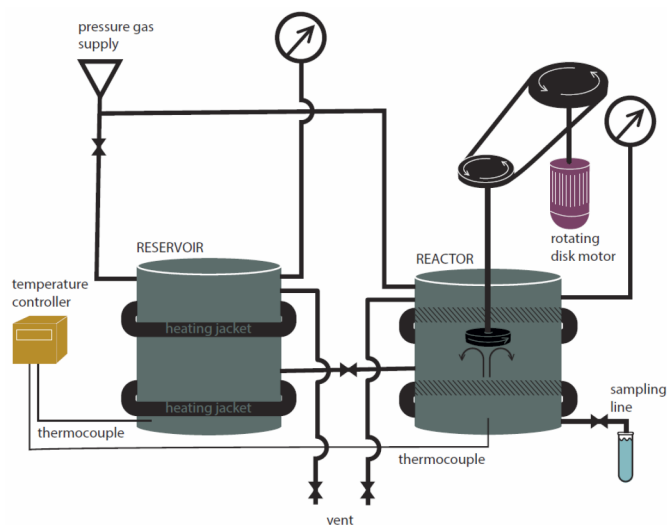


Figure 1—Schematic diagram of the rotating disk apparatus (RDA) [reproduced from reference - 4].

Corrosion Testing

Corrosion tests followed API RP 13B-1 standards, exposing metal coupons to TRA formulations in a sealed reactor at 180°F for 6 hours. The corrosion rate was calculated as:

$$\text{Corrosion Rate} = \frac{\Delta m}{A \cdot t}$$

where Δm is the mass loss (lb), A is the coupon surface area (ft^2), and t is the exposure time (6 hours). Coupons were weighed before and after testing using a high-precision balance (0.1 mg accuracy). Tests were conducted on Cr-13, L-80, P110, and T95 alloys, with no corrosion inhibitors added, to assess the TRA systems' inherent corrosion resistance. Surface analysis via optical microscopy was performed to detect pitting or localized corrosion.

Comparative Analysis

The TRA systems were compared against conventional systems (straight 15% HCl, 5% HCl in-situ gelled acid, and 15% HCl emulsified acid) in terms of dissolution rate, corrosion rate, and operational characteristics (e.g., viscosity, friction pressure). The comparison included data from recent literature to contextualize TRA performance within the current state of acid stimulation technology [12, 13].

Results and Discussion

Emulsified acid

Figure 1a presents a 20X magnified microscopic image of emulsified acid. In this image, the lighter droplets represent the acid-internal phase, while the continuous phase is identified as diesel. Figure 2 illustrates the diameter distribution of the acid droplets within this emulsion. Approximately 50% of the droplets exhibit a diameter of 6.9 μm , with the smallest and largest observed diameters being 1.3 μm and 64.0 μm , respectively. Such emulsions are characterized by a low effective diffusion coefficient ($4.905 \times 10^{-11} \text{ cm}^2/\text{s}$) and significant retardation properties.

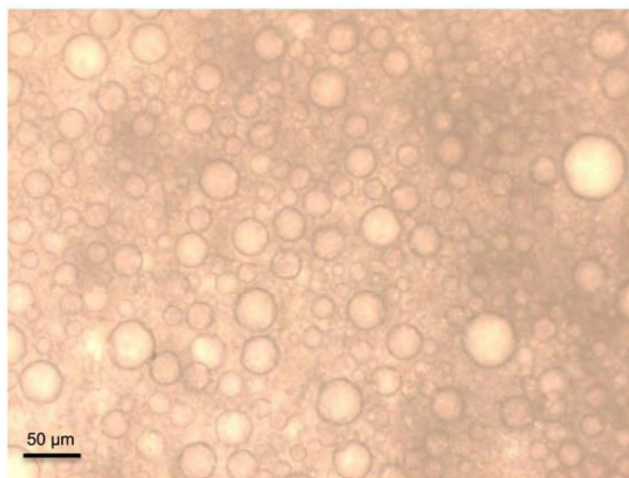


Figure 1a—15 wP/oHCl emulsion micro image taken at MX magnification. Note scale bar of 50 μm [reproduced from reference - 4].

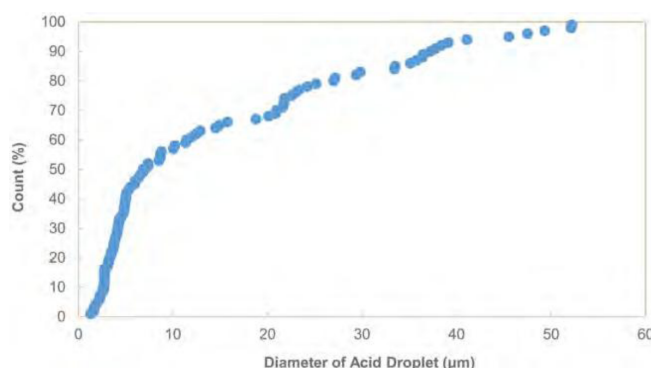


Figure 2—Distribution of acid droplet sizes (diameters) in the emulsified acid used in this study [reproduced from reference - 4].

Dissolution Performance

The dissolution rates of TRA systems and conventional acids at 250°F (except emulsified acid at 275°F) are presented in Table 1.

Table 1—Calcite dissolution rates at 250 °F (except emulsified acid at 275 °F)

Fluid	Temperature (°F)	Calcite Dissolution rate (mol/sec. cm ²)	Ratio
HCl 15%	250	5.77E-05	1.0
HCl 5% (in-situ gelled Acid)		1.93E-05	3.0
HCl 15% (Emulsified Acid)	275	1.60E-05	3.6
Package 1 (with 15% HCl)	250	6.28E-09	9182.0
Package 2 (with 15% HCl)		4.31E-08	1337.9
Package 3 (with 15% HCl)		6.17E-07	93.5
Package 4 (with 15% HCl)		5.41E-06	10.7
Package 5 (with 15% HCl)		1.22E-05	4.7
Package 6 (with 15% HCl)		4.67E-05	1.2
Package 5 (with 15% HCl)	180	2.57E-07	

Straight 15% HCl exhibited a dissolution rate of $5.77\text{E-}05 \text{ mol/sec}^{\wedge}\text{cm}^2$, serving as the baseline. The TRA systems demonstrated a wide range of retardation, with Package 1 achieving a dissolution rate of $6.28\text{E-}09 \text{ mol/sec}^{\wedge}\text{cm}^2$ (retardation factor of 9200) and Package 6 at $4.67\text{E-}05 \text{ mol/sec}^{\wedge}\text{cm}^2$ (retardation factor of 1.2). In contrast, 5% HCl in-situ gelled acid and 15% HCl emulsified acid achieved retardation factors of 3.0 and 3.6, respectively, indicating significantly less control over reaction rates. SEM analysis of post-test calcite disks revealed distinct dissolution patterns. Slower-reacting TRA packages (e.g., Package 1) formed uniform, elongated wormholes, indicative of controlled acid spending and deeper penetration potential (see [Figure 3](#)). Faster-reacting packages (e.g., Package 6) produced broader, less defined dissolution patterns, suitable for near-wellbore stimulation. These findings align with Zhou et al. [13], who noted that controlled reaction rates enhance wormhole efficiency in high-temperature reservoirs. [Figure 4](#) shows that different disc shapes show different reaction rates, which causes etching of the rock.

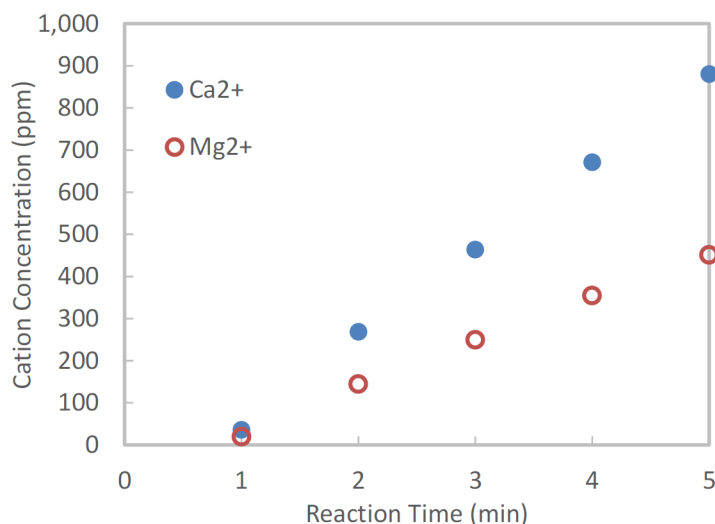
**Figure 3—Dissolution rate (mole/s/cm²) Vs reaction time (min).**



Figure 4—Calcite marble disks after RDA test.

Corrosion Performance

Corrosion rates for TRA systems on Cr-13 steel at 250 °F are shown in Table 2. All packages exhibited rates below the industry threshold of 0.05 lb/ft² in 6 hours, ranging from 0.00001 lb/ft² (Packages 2 and 5) to 0.05606 lb/ft² (Package 6). Tests on L-80, P110, and T95 alloys confirmed similarly low rates, with no evidence of pitting or localized corrosion. The absence of corrosion inhibitors highlights the TRA systems' intrinsic corrosion resistance, a significant improvement over conventional HCl systems that require intensive additive packages [13].

Table 2—Corrosion rates on Cr-13 steel at 250°F and 180 °F (6-hour exposure)

Fluid	Temperature (°F)	Cr-13 Corrosion Rate (lb/ft ² in 6h)
Package 1	250	0.00041
Package 2		0.00001
Package 3		0.00002
Package 4		0.04376
Package 5		0.00001
Package 6		0.05606
Package 5	180	0.00001

Comparison with Conventional Systems

Compared to emulsified and in-situ gelled acids, TRA systems offer superior performance in several areas. Their single-phase, Newtonian behavior eliminates the high friction pressures of emulsified acids, enabling higher injection rates and simpler pumping operations [14]. The absence of diesel or polymer residues reduces environmental impact and formation damage risk, addressing concerns raised by [12]. The TRA systems' tunability—reaction rates adjustable from 1.2 to 9000 times slower than 15% HCl, compatibility with acid concentrations up to 32%, and operation across 150-350°F—provides unmatched flexibility for diverse reservoir conditions.

The TRA systems' ease of preparation and compatibility with batch or continuous injection streamline field operations, reducing logistical complexity compared to emulsified acids. Their low corrosion rates eliminate the need for costly inhibitors, lowering operational costs and environmental footprint. The ability to tailor viscosity with viscosifiers supports applications in both matrix acidizing and acid fracturing, aligning with industry trends toward versatile, sustainable stimulation fluids [13]. While the TRA systems demonstrate significant advantages, challenges remain in scaling up to field applications. Ongoing field trials are evaluating their performance in real-world conditions, particularly in high-temperature,

heterogeneous reservoirs. Future research will focus on optimizing retarder formulations for ultra-high-temperature (>350°F) applications and exploring compatibility with complex carbonate mineralogies (e.g., dolomite-dominated reservoirs).

Conclusions

This study introduces a novel class of Tunable Retarded Acid (TRA) systems that can revolutionize carbonate reservoir stimulation. Laboratory tests demonstrate their ability to achieve highly tunable reaction rates (1.2 to 9000 times slower than 15% HCl), low corrosion rates without inhibitors, a wide temperature range (150 - 350°F), and Newtonian flow behavior, surpassing the performance of conventional emulsified and in-situ gelled acids. The TRA systems' compatibility with diverse acid types, temperatures, and injection methods, combined with their environmental and operational benefits, positions them as a transformative solution for matrix acidizing and acid fracturing. This TRA system works with both organic and inorganic acid systems.

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